

yumeng ma

HUMAN-AI INTERACTION ACCESSIBLE COMPUTING SYSTEM DESIGN & EVALUATION

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Education

University of Washington

Seattle, WA

PHD IN COMPUTER SCIENCE AND ENGINEERING

2024 - Present

GPA: 4.0 / 4.0

Advisor: Jacob O. Wobbrock

Brown University

Providence, RI

Sc.M IN COMPUTER SCIENCE

2022 - 2024

GPA: 4.0 / 4.0

Advisor: Jeff Huang

Arizona State University

Tempe, AZ

B.S. IN COMPUTER SCIENCE AND PSYCHOLOGY

2019 - 2022

Barrett Honors College

GPA: 4.0 / 4.0 Summa Cum Laude

Thesis: "Assessing Exhibit Interaction through Bridging the Connections Between Biometrics and Traditional Tools"

Publications

* denotes equal contribution, † acceptance rates appear in superscript brackets when available

CONFERENCE PROCEEDINGS

- [C.4] **Ma, Y.**, Wobbrock, J. O., Devon, J., Gustafson, A., Fairchild, M., Mankoff, J., and Potluri, V. (2026). Prompt and circumstance: Evaluating accessibility in LLM-generated HTML form components. *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26)*. Porto, Portugal (October 25–28, 2026). New York, NY: ACM Press. *To appear.* ^[30%]
- [C.3] **Ma, Y.**, Hiniker, A., and Wobbrock, J. O. (2026). Quantifying the novelty bias when evaluating interactive prototypes. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '26)*. Barcelona, Spain (April 13–17, 2026). New York, NY: ACM Press. Article No. 1011. **Honorable Mention Award.** *paper* ^[25%]
- [C.2] Iftikhar, Z., **Ma, Y.**, and Huang, J. (2023). "Together but not together": Evaluating typing indicators for interaction-rich communication. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '23)*. Hamburg, Germany (April 23–28, 2023). New York, NY: ACM Press. Article No. 724. *paper* ^[28%]
- [C.1] Gonsheer, I., **Ma, Y.**, Pineda-Dominguez, I., Lee, M., and Han, Y. (2023). The mixed reality passthrough window: Rethinking the laptop videoconferencing experience. *Proceedings of the 9th International Conference on Human Interaction and Emerging Technologies: Artificial Intelligence and Future Applications (IHET-AI '23)*. Lausanne, Switzerland (April 21–23, 2023). AHFE International. *paper*

EXTENDED ABSTRACTS & WORKS-IN-PROGRESS

- [P.2] **Ma, Y.** and Malhotra, R. (2026). Designing graceful degradation for multimodal autonomous vehicle cabins. *Adjunct Proceedings of the 18th International ACM Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26)*. Gothenburg, Sweden (September 20–23, 2026). New York, NY: ACM Press. *paper*
- [P.1] **Ma, Y.*** and Ren, J.* (2023). ProactiveAgent: Personalized context-aware reminder system. *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*. San Francisco, CA, USA (October 29–November 1, 2023). New York, NY: ACM Press. Article No. 115. *paper*

Research Experience

UW ACE Lab Advisor: Jacob O. Wobbrock

Seattle, WA

GRADUATE RESEARCH ASSISTANT

2024–Present

- Conducting research on system evaluation and accessible computing through studies on novelty bias in interactive prototype evaluation [C.3] and accessibility benchmarking for LLM-generated web interfaces in collaboration with Microsoft [C.4].

Brown Department of Neuroscience Advisor: Michael Paradiso

Providence, RI

COMPUTER VISION RESEARCH ASSISTANT

2023–2024

- Built a visual prosthesis system that helps blind and low vision users locate everyday objects and recognize nearby text.

UCLA HCI Lab Advisor: Xiang “Anthony” Chen

Los Angeles, CA

HCI RESEARCH INTERN

Summer 2023

- Led development of a proactive reminder system that uses personal context, interaction history, spoken input, and large language models. [P.1]

Brown Department of Engineering Advisor: Ian Gonsher

Providence, RI

DESIGN RESEARCH ASSISTANT

2022–2023

- Developed an augmented-reality display concept that supports interaction between colocated and remote participants during videoconferencing. [C.1]

Brown HCI Lab Advisor: Jeff Huang

Providence, RI

DISTRIBUTED RESEARCH EXPERIENCES FOR UNDERGRADUATES FELLOW

Summer 2022

- Designed novel typing indicators and conducted mixed-method analysis linking them with stronger perceived co-presence. [C.2]

ASU SoLET Lab Advisor: Danielle McNamara

Tempe, AZ

PSYCHOLOGY RESEARCH ASSISTANT

2022

- Tested and evaluated intelligent tutoring systems designed to support learning across underrepresented student groups.

NASA Psyche Mission Advisor: Maria Elena Chavez Echeagaray

Tempe, AZ

HCI RESEARCH ASSISTANT

2021–2022

- Developed an interactive Psyche spacecraft web application and led user studies using biometric measures to assess public engagement.

Work Experience

Infosys Applied Research Center for Autonomous Machines Mentor: Rani Malhotra

RESEARCH SCIENTIST INTERN

Fall 2026 – present

- Developing adaptive human–vehicle interaction concepts for autonomous mobility, with in-vehicle AI that learns passenger preferences across repeated rides.

Amazon Devices & Services Mentors: Anna Hernandez, Ronak Mohanty

UX RESEARCH INTERN, HUMAN FACTORS TEAM

Summer 2026

- Led end-to-end research on flash-suppression experiences for photosensitive media viewers, resulting in seven evaluated interventions and product recommendations.

Infosys Applied Research Center for Autonomous Machines Mentor: Rani Malhotra

RESEARCH SCIENTIST INTERN

Summer 2025

- Conducted design research on multimodal autonomous vehicle interfaces and developed graceful degradation strategies through hazard escalation storyboards and user studies. [P.2]

Infosys Living Labs

UX DESIGN INTERN

Summer 2024

- Led the design of an accessible phygital interface that used emerging AI capabilities to support retail experiences across varied user needs.

Brown University + NASA

Providence, RI

SOFTWARE DEVELOPER

2023–2024

- Led development of a cross-platform mobile application using sound and haptics to help blind and low vision users explore and learn about space.

Civillience

PROJECT MANAGEMENT & UX DESIGN INTERN

2023

- Led UI/UX design for a web application with interactive health data visualizations and built reusable components for a consistent product design system.

Awards, Fellowships, & Grants

- 2025 **Support for “AI Model Accessibility Checker”**, *Microsoft*
- 2024 **NSF Graduate Research Fellowship**, *National Science Foundation* (three years of support)
- 2024 **Paul G. Allen School Fellowship**, *University of Washington*
- 2019 **Microsoft DisAbility Scholarship**, *Microsoft* (four years of support)

Invited Talks

- 2026 *Panelist*, Microsoft Ability Summit *Redmond, WA*
- 2025 *Lightning Talk*, Open Source Accessibility Summit *Raleigh, NC*

Teaching

- Spring '26 **CSE 440 Human-Computer Interaction**, *Graduate TA for Amy Zhang, UW*
- Winter '26 **CSE 443 Digital Accessibility**, *Graduate TA for Jennifer Mankoff, UW*
- Spring '22 **CSE 365 Information Assurance**, *TA for Adam Doupé and Tiffany Bao, ASU*

Service & Professional Activities

- 2026 **Reviewer**, ACM UIST & ACM AutoUI
- 2022–2024 **Master’s Student Representative**, Brown University CS Diversity Committee *Providence, RI*
- 2023 **Student Volunteer**, ACM UIST *San Francisco, CA*